



WAVE MOTION — FULL MIND MAP

तरंग गति · Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Mechanical, Longitudinal, Transverse & Stationary Waves
2. Radio Propagation, Ionosphere & Waves in Vacuum
3. Cosmic Rays, Radiation, Albedo & Atmospheric Heating
4. Electromagnetic Spectrum — Speed, Frequency, Energy & X-rays
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7. Ionosphere Communication & Rain Fade
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1. Mechanical, Longitudinal, Transverse & Stationary Waves

- Sound ek mechanical wave hai; air/fluid me particles propagation direction ke parallel vibrate karte hain, isliye sound longitudinal mechanical wave hoti hai. Radio, X-ray aur light electromagnetic waves hain. [Q1–Q3]
- Longitudinal wave me do consecutive compressions ya rarefactions ke beech ki distance wavelength (λ) hoti hai; transverse wave me consecutive crests/troughs ke beech ki distance bhi wavelength hoti hai. [Q4, Q20]
- Sitar ki stretched string pluck karne par transverse waves reflect hokar stationary transverse wave pattern banati hain; instrument body bhi forced vibration se sound radiate karti hai. [Q5]
- Sound vacuum me travel nahi kar sakti because mechanical wave ko particle medium chahiye; solid, liquid aur gas me propagate kar sakti hai. [Q10]

2. Radio Propagation, Ionosphere & Waves in Vacuum

- Long-distance radio communication me ionosphere important hai: upper atmosphere ka ionized region radio propagation ko affect/return kar sakta hai; source questions long radio waves/wireless deflection-reflection ko ionosphere se link karte hain. [Q6–Q8]
- UHF roughly 300 MHz–3 GHz range hai; high-frequency UHF signals commonly line-of-sight/space-wave propagation use karte hain, ground-wave ya ordinary ionospheric sky-wave route nahi. [Q9]
- Radio waves electromagnetic hain aur uncharged radiation hone ke karan ordinary electric/magnetic fields se beam-path deflection nahi dikhati. [Q11]
- Electromagnetic radiation ko material medium ki requirement nahi hoti; conduction/convection ke unlike radiation vacuum se pass kar sakti hai. [Q14]



EXAM TRAP — Q6–Q9 — Radio Propagation Depends on Frequency

- ▶ Ionosphere kuch radio-frequency bands ko refract/return kar sakta hai, but sab radio waves same tarah reflect nahi hoti.
- ▶ UHF generally line-of-sight/space-wave mode me propagate karta hai; exam statement ko frequency band ke context me padho.

3. Cosmic Rays, Radiation, Albedo & Atmospheric Heating

- Cosmic rays electromagnetic waves nahi, mostly very-high-energy charged particles hain jo Sun aur extra-solar/galactic sources se aa sakte hain. [Q12–Q13]
- Albedo kisi surface/body ki incident radiation ko reflect karne ki fraction/reflecting power hai; fresh snow high albedo aur dark/water surfaces comparatively low albedo show kar sakte hain. [Q15]
- Earth incoming solar shortwave energy absorb karke longwave terrestrial infrared radiation emit karti hai; CO_2 aur other greenhouse gases is longwave radiation ko absorb karte hain, isliye atmosphere largely terrestrial radiation se heat hota hai. [Q16–Q18]
- Earth surface absorbed energy ko ultraviolet nahi, mainly infrared/longwave form me re-radiate karti hai; clouds aur atmospheric gases incoming solar radiation ka significant fraction reflect/absorb bhi karte hain. [Q19]

4. Electromagnetic Spectrum — Speed, Frequency, Energy & X-rays

- Vacuum me sab electromagnetic waves ki speed $c \approx 3 \times 10^8$ m/s hoti hai. [Q21]
- X-rays electrically neutral electromagnetic radiation hain, isliye electric ya magnetic field se deflect nahi hoti; matter/crystals se diffraction/scattering ho sakti hai. [Q22]

- Supersonic “wave” EM spectrum ka member nahi; term speed-of-sound se faster motion ko describe karta hai. Gamma, X-ray aur UV electromagnetic radiation hain. [Q23]
- EM spectrum wavelength order (long → short): Radio > Microwave > Infrared > Visible > Ultraviolet > X-ray > Gamma. Frequency/energy ka order reverse hota hai. [Q24–Q26]
- Microwaves ko broad definitions me roughly 300 MHz–300 GHz tak maana ja sakta hai; source option 1–300 GHz ko most appropriate choose karta hai. [Q27]

⚠ **EXAM TRAP — Q27 — Microwave Frequency Range**

- ▶ Microwave band ki exact lower boundary source/standard par vary karti hai. Broad engineering usage ≈ 300 MHz–300 GHz bhi milti hai.
- ▶ HTML source options me 1–300 GHz answer chosen hai; exam me given options follow karo.

5. EM-wave Applications — Radar, Remote Sensing, IR & Medical Use

- Crystal structure determine karne me X-ray diffraction widely used hai because X-ray wavelength atomic spacing ke comparable hoti hai. [Q28]
- RADAR radio/microwave-frequency electromagnetic waves se target aircraft/objects ki range, direction ya speed infer karta hai. [Q29]
- Remote sensing Earth/objects se reflected ya emitted electromagnetic radiation ko distance se measure karta hai; sensors spectrum ke different bands use kar sakte hain. [Q30]
- Ozone depletion se shorter-wavelength UV surface tak pahunch sakti hai; source calculation minimum ≈ 100 nm = 10^{-7} m ko option ke roop me leta hai. [Q31]
- Night-vision systems infrared radiation detect/use karte hain. Common TV remotes bhi typically infrared use karte hain. [Q32–Q35]
- Q36 ke options me infrared absent hai; source “radio waves” choose karta hai because RF remotes bhi exist karte hain, lekin conventional TV remote ka standard exam answer generally infrared hota hai. [Q36]

⚠ **EXAM TRAP — Q36 — TV Remote: Infrared vs Radio**

- ▶ Conventional TV remotes ka standard answer infrared (IR) hai—Q32–Q35 bhi isi concept ko test karte hain.
- ▶ Q36 me IR option hi nahi diya gaya; source RF remote existence ke basis par “radio waves” accept karta hai. Isko general rule mat samajhna.

6. Telecommunication, Microwave, Modulation & FM

- Point-to-point aur satellite telecommunication me microwaves widely used hoti hain because high carrier frequency large information capacity aur directional links allow karti hai. [Q37]
- Carrier radio wave par information signal superimpose karne ki process modulation kehlati hai. [Q38]
- FM me carrier frequency information signal ke according vary karti hai; FM broadcasting AM se higher frequency band aur wider bandwidth use karti hai. “FM frequency range lower than AM” false hai. [Q39]
- Source medium-speed 1200–2400 bit/s communication ke liye frequency modulation ko preferred batata hai; ye historical/source-specific data-communication framing hai. [Q40]
- Standard FM broadcast band 88–108 MHz hai. [Q41]
- Wave relation $f = c/\lambda$. $\lambda = 30$ m aur $c = 3 \times 10^8$ m/s se $f = 10^7$ Hz = 10 MHz. [Q42]

⚠ **EXAM TRAP — Q40 — 1200–2400 bit/s FM Claim**

- ▶ Ye source ka older communication/modem-style statement hai, modern digital communication ka universal rule nahi.
- ▶ Exam-source mapping preserve ki gayi hai: source answer Frequency Modulation hai.

7. Ionosphere Communication & Rain Fade

- Ionosphere radio communication ko charged ions/electrons ki wajah se facilitate karta hai—not ozone. “Radio waves have long wavelength” generally true ho sakta hai, but given reason ke roop me sufficient nahi; source Q43 me neither statement accepted hai. [Q43]
- Rain satellite/microwave signals ko absorb/scatter karke attenuation (“rain fade”) cause kar sakti hai; weakened signal small receiving dish ke liye threshold ke below ja sakta hai. [Q44]

EXAM TRAP — Q43–Q44 — Ionosphere & Rain Fade

- ▶ Ionospheric radio return ionized particles/electron-density profile se related hai, ozone reflection se nahi.
- ▶ Rain fade especially microwave/satellite links me absorption + scattering se hota hai; dish size signal margin ko affect kar sakta hai but rain ka root cause antenna size nahi.

8. X-rays in Diagnosis & Wilhelm Röntgen

- X-rays medical imaging me use hoti hain; source intestinal-disease diagnosis ko X-rays se link karta hai. [Q45]
- CT scan multiple X-ray projections + computer reconstruction se cross-sectional images banata hai. [Q46]
- Wilhelm Conrad Röntgen ne X-rays discover ki thi; source MCQ option “X-Ray Machine” ko accepted answer deta hai. [Q47]

EXAM TRAP — Q47 — Discovery vs “X-Ray Machine”

- ▶ Historically Wilhelm Conrad Röntgen is credited with discovering X-rays (1895).
- ▶ Source MCQ wording/options “X-Ray Machine” ko accepted answer dete hain; factual concept ko discovery of X-rays ke roop me yaad rakho.

✓ **Source coverage: Q1–Q47**